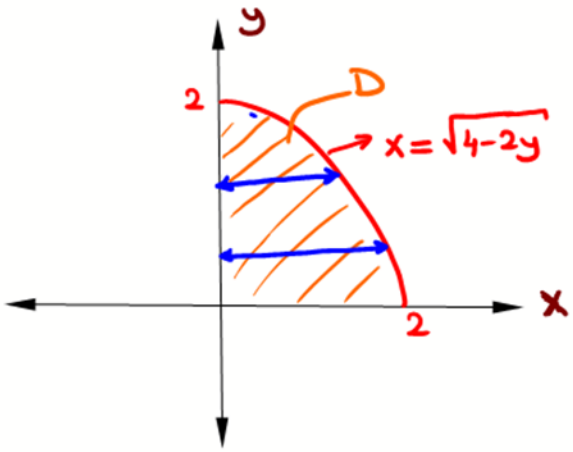
1. D birinci çeyrek bölgede $x^2 = 4-2y$ eğrisi ile sınırlı kapalı bir bölge olduğuna göre bu bölge üzerinde $\iint_D \frac{1}{\sqrt{2y-y^2}}\,dA$ integralinin değeri aşağıdakilerden hangisidir?

a) 1 b) 4 c) -4 d) -1 e) $\frac{3}{2}$



$$\iint_D \frac{1}{\sqrt{2y-y^2}}\,dA = \int_{y=0}^{y=2}\int_0^{\sqrt{4-2y}} \frac{1}{\sqrt{y}\sqrt{2-y}}\,dx\,dy$$

$y(2-y)$

$$= \int_0^2 x\Big|_0^{\sqrt{4-2y}} \frac{1}{\sqrt{y}\sqrt{2-y}}\,dy$$

$$= \int_0^2 \sqrt{2}\cdot\cancel{\sqrt{2-y}}\,\frac{1}{\sqrt{y}\cdot\cancel{\sqrt{2-y}}}\,dy = \int_0^2 \sqrt{2}\,\frac{dy}{\sqrt{y}}$$

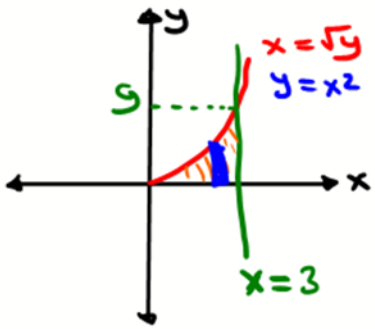
$$= \sqrt{2}\cdot 2\sqrt{y}\Big|_0^2 = 2\sqrt{2}\cdot\sqrt{2} = 4$$

B şıkkı

4. $\int_0^9 \int_{\sqrt{y}}^3 e^{x^3}\,dx\,dy = ?$ → integral bu haliyle çözülemez! Sıralama değişmeli!

a) $\frac{1}{3}(e^9-1)$ b) $\frac{1}{2}(e^{27}-1)$ c) $\frac{1}{3}(e^{27}-1)$ d) $\frac{1}{3}(e^3-1)$ e) $\frac{1}{3}(e^{27}+1)$

$y=0 \quad x=\sqrt{y}$
$y=9 \quad x=3$



$dx\,dy \rightarrow dy\,dx$ olmalı!

$$\int_0^3 \int_0^{x^2} e^{x^3}\,dy\,dx$$

$$= \int_0^3 y\Big|_0^{x^2} e^{x^3}\,dx = \int_0^3 x^2 e^{x^3}\,dx$$

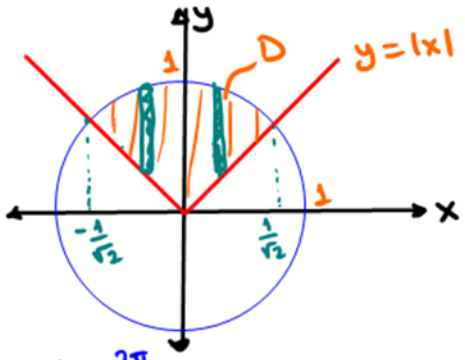
$3x^2$

$$= \frac{1}{3}e^{x^3}\Big|_0^3 = \frac{1}{3}(e^{27}-1)$$

C şıkkı

8. D, üstten $x^2+y^2=1$ egrisi ve alttan $y=|x|$ ile sınırlı kapalı bir bölge olduğuna göre bu bölge üzerinde $\iint_D y\,dA$ integralinin değeri aşağıdakilerden hangisidir?

a) $\frac{\sqrt{2}}{3}-1$ b) $\frac{\sqrt{2}}{3}$ c) $-\frac{\sqrt{2}}{3}$ d) $\frac{2\sqrt{2}}{3}$ e) $\frac{\sqrt{2}}{2}$



1. yol

$$\int_{-\frac{1}{\sqrt{2}}}^{0}\int_{-x}^{\sqrt{1-x^2}} y\,dy\,dx + \int_{0}^{\frac{1}{\sqrt{2}}}\int_{x}^{\sqrt{1-x^2}} y\,dy\,dx$$

$$= \int_{-\frac{1}{\sqrt{2}}}^{0} \frac{y^2}{2}\Big|_{-x}^{\sqrt{1-x^2}}dx + \int_{0}^{1/\sqrt{2}} \frac{y^2}{2}\Big|_{x}^{\sqrt{1-x^2}}dx$$

$$= \frac{1}{2}\int_{-\frac{1}{\sqrt{2}}}^{0}\left[(1-x^2)-x^2\right]dx + \frac{1}{2}\int_{0}^{1/\sqrt{2}}(1-x^2-x^2)\,dx$$

$$= \frac{1}{2}\left(x-\frac{2}{3}x^3\right)\Big|_{-\frac{1}{\sqrt{2}}}^{\frac{1}{\sqrt{2}}} = 2\cdot\frac{1}{2}\left(\frac{1}{\sqrt{2}}-\frac{2}{3}\frac{1}{2\sqrt{2}}\right) = \frac{\sqrt{2}}{3}$$

B şıkkı

2. yol

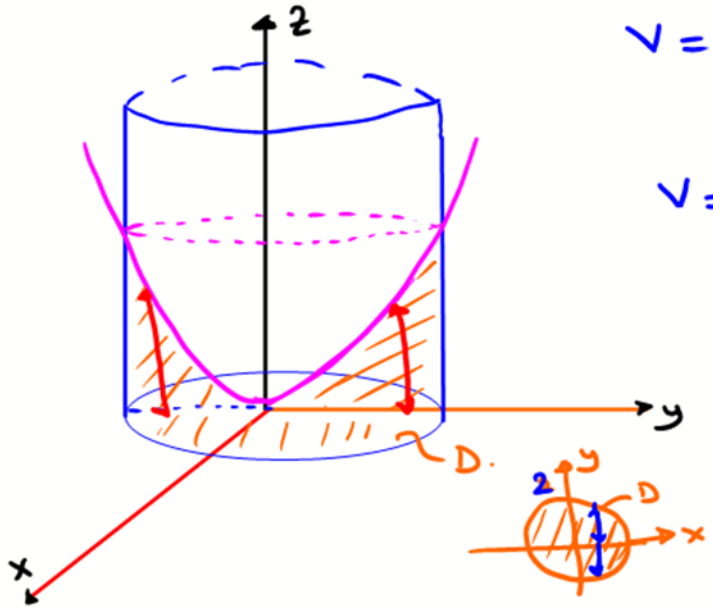
$$\int_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\int_{0}^{1} r\sin\theta\cdot r\,dr\,d\theta$$

$$= -\cos\theta\Big|_{\frac{\pi}{4}}^{\frac{3\pi}{4}}\cdot\frac{r^3}{3}\Big|_0^1 = \left(\frac{1}{\sqrt{2}}+\frac{1}{\sqrt{2}}\right)\cdot\frac{1}{3} = \sqrt{2}/3$$

9. $x^2+y^2=4$ silindirinin içinde $z=x^2+y^2$ paraboloidinin altında $z=0$ düzleminin üstünde oluşan cismin hacmini veren katlı integral aşağıdakilerden hangisidir?

a) $\int_0^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}(x^2+y^2)\,dy\,dx$ b) $\int_{-2}^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}(x^2+y^2)\,dy\,dx$

c) $\int_0^2\int_0^{\sqrt{4-y^2}}(x^2+y^2)\,dx\,dy$ d) $\int_0^2\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}(x^2+y^2)\,dy\,dx$ e) $\int_{-2}^2\int_{-\sqrt{4-y^2}}^{\sqrt{4-y^2}}(x^2+y^2)\,dy\,dx$



$$V = \iint_D (x^2+y^2-0)\,dA$$

$$V = \int_{-2}^{2}\int_{-\sqrt{4-x^2}}^{\sqrt{4-x^2}}(x^2+y^2)\,dy\,dx$$

B şıkkı

11. $\int_0^1 \int_{\sqrt{y}}^1 \cos x^3 \, dx dy = ?$ → bu haliyle çözülemez! sırasını değiştirmeliyim! dxdy → dydx olmalı.

a) $\frac{1}{3}\cos 1$ b) $\frac{1}{2}\sin 1$ c) $\frac{1}{3}\sin 1$ d) $\sin 1$ e) $\sin \frac{1}{3}$

$x = \sqrt{y}$ $\quad y = 0$
$x = 1$ $\quad y = 1$
$y = x^2$



$$\int_0^1 \int_0^{x^2} \cos x^3 \, dy dx = \int_0^1 y \Big|_0^{x^2} \cdot \cos x^3 \, dx$$

$$= \int_0^1 \frac{3}{3} x^2 \cdot \cos x^3 \, dx = \frac{1}{3} \sin x^3 \Big|_0^1$$

$$= \frac{1}{3}\sin 1$$

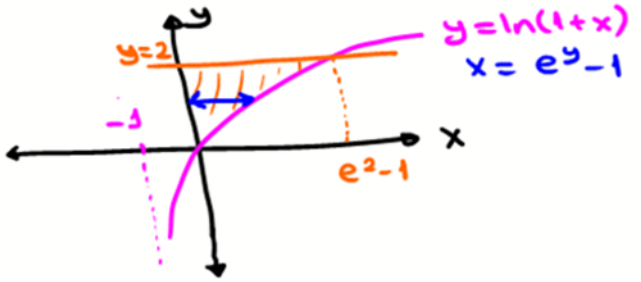
C şıkkı

18. $\int_0^{e^2-1} \int_{\ln(1+x)}^2 f(x,y) \, dy dx$ integralinde integrasyon sırası değiştirildiğinde aşağıdaki integrallerden hangisi elde edilir?

a) $\int_0^2 \int_0^{e^y-1} f(x,y) dx dy$ b) $\int_0^1 \int_0^{e^y-1} f(x,y) dx dy$ c) $\int_{-2}^2 \int_0^{e^y-2} f(x,y) dx dy$

d) $\int_0^2 \int_1^{e^y-1} f(x,y) dx dy$ e) $\int_0^2 \int_0^{e^y-2} f(x,y) dx dy$

$y = 2$ $\quad x = 0$
$y = \ln(1+x)$ $\quad x = e^2 - 1$

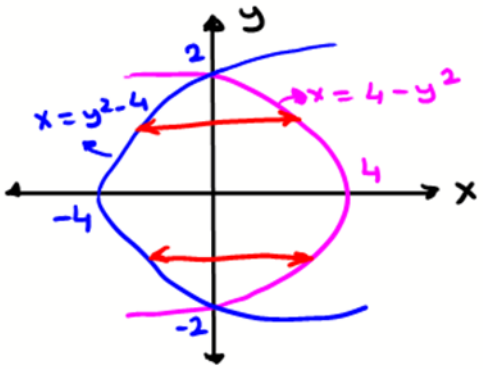


$$\int_{y=0}^{y=2} \int_0^{e^y-1} f(x,y) \, dx dy$$

// A şıkkı

20. $x = 4-y^2$ ve $x = y^2-4$ eğrileri ile sınırlı düzlem bölgenin alanını veren integral aşağıdakilerden hangisidir?

a) $\int_{-2}^{2}\int_{4-y^2}^{y^2-4} dxdy$ b) $\int_{-2}^{2}\int_{4-y^2}^{y^2-4} dydx$ c) $\int_{0}^{4}\int_{4-y^2}^{y^2-4} dydx$ d) $\int_{-2}^{2}\int_{y^2-4}^{4-y^2} dydx$ e) $\int_{-2}^{2}\int_{y^2-4}^{4-y^2} dxdy$



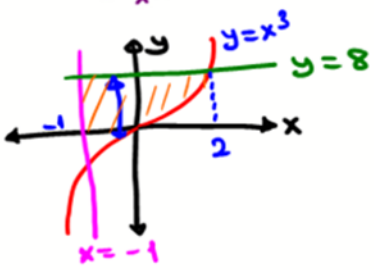
↓ sıklardaki integraller tek parçalı o yüzden bölge dxdy ile hesaplanmış

$$A = \int_{y=-2}^{y=2}\int_{y^2-4}^{4-y^2} dxdy$$

E sıkkı

24. D bölgesi ; $y = x^3$ eğrisi ile $y=8$ ve $x=-1$ doğruları ile sınırlı bölge olmak üzere, D bölgesinin alanını veren iki katlı integral aşağıdakilerden hangisidir?

a) $\int_{-1}^{2}\int_{x^3}^{-1} dydx$ b) $\int_{-1}^{2}\int_{-8}^{x^3} dydx$ c) $\int_{-1}^{2}\int_{\sqrt[3]{y}}^{8} dydx$ d) $\int_{-1}^{2}\int_{x^3}^{8} dydx$ e) $\int_{-1}^{2}\int_{x^3}^{0} dydx$

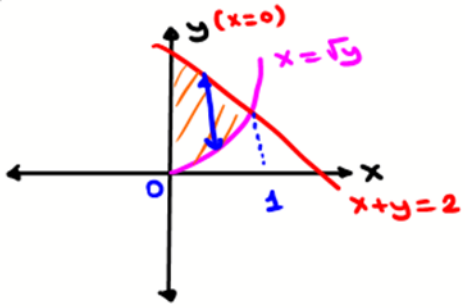


$$A = \int_{x=-1}^{x=2}\int_{x^3}^{8} dydx$$

D sıkkı

26. R bölgesi ; $x = \sqrt{y}$ eğrisi ve $x+y=2$; $x=0$ doğruları ile sınırlanmış bölge olsun. R bölgesinin alanını veren integral aşağıdakilerden hangisidir?

a) $\int_{0}^{1}\int_{\sqrt{x}}^{2-x} dxdy$ b) $\int_{0}^{1}\int_{y^2}^{2-y} dydx$ c) $\int_{0}^{1}\int_{x^2}^{2-x} dydx$ d) $\int_{0}^{1}\int_{\sqrt{y}}^{2-y} dxdy$ e) $\int_{0}^{1}\int_{2-x}^{x^2} dydx$



$$A = \int_{x=0}^{x=1}\int_{x^2}^{2-x} dydx$$

C şıkkı

25. $\int_0^{\frac{2}{3}} \int_{2x}^{2-x} f(x,y)\,dydx$ integralini integrasyon sırasını değiştirerek yeniden yazdığımızda elde edilen integral aşağıdakilerden hangisidir?

a) $\int_0^{\frac{2}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{2}{3}}^{2} \int_{\frac{y}{2}}^{2-y} f(x,y)\,dxdy$

b) $\int_{2x}^{2-x} \int_0^{\frac{2}{3}} dxdy$

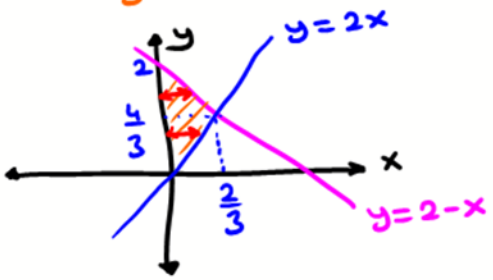
c) $\int_0^{\frac{4}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{4}{3}}^{2} \int_{\frac{y}{2}}^{2-y} f(x,y)\,dxdy$

d) $\int_0^{\frac{2}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_0^{\frac{4}{3}} \int_0^{2-y} f(x,y)\,dxdy$

e) $\int_0^{\frac{4}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{4}{3}}^{2} \int_0^{2-y} f(x,y)\,dxdy$

$x=0 \quad y=2x$

$x=\frac{2}{3} \quad y=2-x$



$$\int_{y=0}^{\frac{4}{3}} \int_0^{\frac{y}{2}} f(x,y)\,dxdy + \int_{\frac{4}{3}}^{2} \int_0^{2-y} f(x,y)\,dxdy$$

E şıkkı

29. $\int_{-3}^{1-e} \int_{1-x}^{4} f(x,y)\,dydx + \int_{1-e}^{1} \int_{e}^{4} f(x,y)\,dydx + \int_{1}^{\ln 4} \int_{e^x}^{4} f(x,y)\,dydx = ?$

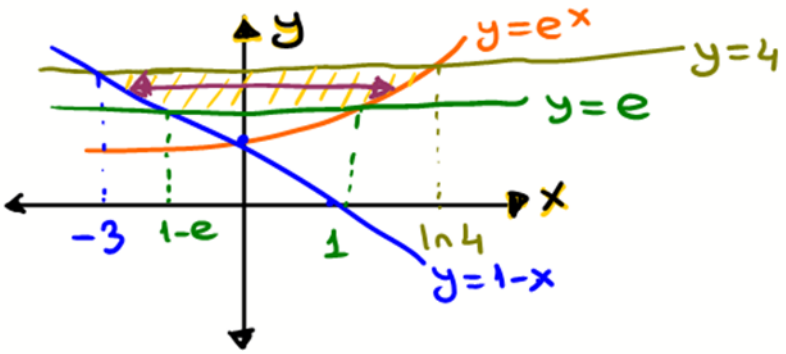
a) $\int_e^4 \int_{1-y}^{\ln y} f(x,y)\,dxdy$

b) $\int_e^4 \int_{1+y}^{\ln y} f(x,y)\,dxdy$

c) $\int_0^1 \int_e^{e^y} f(x,y)\,dxdy$

d) $\int_{-3}^{1} \int_{1+y}^{e^y} f(x,y)\,dxdy$

e) $\int_{-3}^{1} \int_{e^y}^{4} f(x,y)\,dxdy$



$$\int_{y=e}^{y=4} \int_{x=1-y}^{x=\ln y} f(x,y)\,dxdy$$

A şıkkı

30. f, diferansiyellenebilir bir çift fonksiyon olmak üzere, $\int_0^1 f(y)\,dy = 2$ olsun. $\int_{-1}^{1}\int_{0}^{1} f(y)\,x e^{\frac{x^2}{2}}\,dx\,dy = ?$

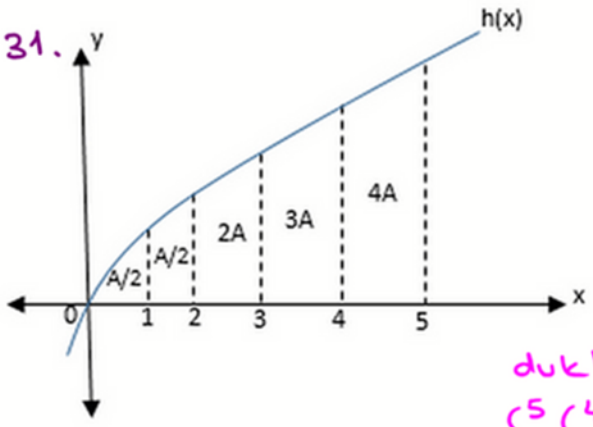
a) $2\sqrt{e}-2$ b) $2e-1$ c) $4\sqrt{e}-4$ d) $4e-1$ e) 0

$$\int_{-1}^{1}\int_{0}^{1} f(y)\,x e^{x^2/2}\,dx\,dy = \int_{-1}^{1} e^{x^2/2}\,dx \Big|_0^1 \int f(y)\,dy$$

$$= \int_{-1}^{1} (\sqrt{e}-1)\,f(y)\,dy = (\sqrt{e}-1)\,2\cdot \underbrace{\int_0^1 f(y)\,dx}_{2} = 4\sqrt{e}-4$$

$f(y)$: çift fonk.

C şıkkı



$f(x,y)$ ve $h(x)$ diferansiyellenebilir fonksiyonlar olmak üzere $\int_1^5 f(x,y)\,dy = h(x)$ ve $\int_0^4 h(x)\,dx = 12$ olsun. Yandaki grafikte, $A/2$, $2A$, $3A$, $4A$ bulundukları bölgenin alanlarını temsil etmektedir. $\int_1^5\int_2^4 f(x,y)\,dx\,dy = ?$

a) 9 b) 10 c) 18 d) 19 e) 20

$$\int_0^4 h(x)\,dx = 3A + 2A + \frac{A}{2} + \frac{A}{2} = 6A = 12 \Rightarrow A = 2br^2$$

$$\int_1^5\int_2^4 f(x,y)\,dx\,dy = \int_2^4 \underbrace{\int_1^5 f(x,y)\,dy}_{h(x)}\,dx = \int_2^4 h(x)\,dx = 5A$$

$$= 5\cdot 2 = 10br^2$$

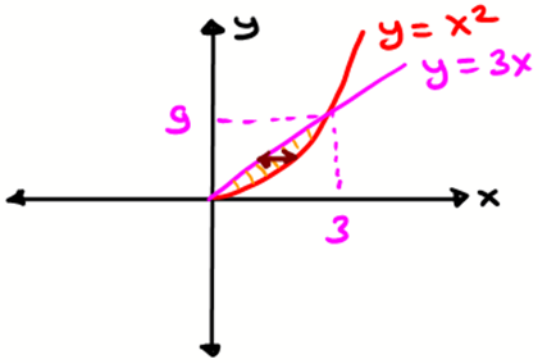
B şıkkı

32. $\int_0^3 \int_{x^2}^{3x} f(x,y)\,dydx$ integralinde integrasyon sırası değiştirildiğinde elde edilen integral aşağıdakilerden hangisidir?

a) $\int_0^3 \int_y^{\sqrt{y}} f(x,y)\,dxdy$ b) $\int_0^{16} \int_{\frac{y}{4}}^{\sqrt{y}} f(x,y)\,dxdy$ c) $\int_0^{25} \int_{\frac{y}{5}}^{\sqrt{y}} f(x,y)\,dxdy$

d) $\int_0^4 \int_{\frac{y}{2}}^{\sqrt{y}} f(x,y)\,dxdy$ e) $\int_0^9 \int_{\frac{y}{3}}^{\sqrt{y}} f(x,y)\,dxdy$

$x=0 \quad y=x^2$
$x=3 \quad y=3x$



$\int_{y=0}^{y=9} \int_{\frac{y}{3}}^{\sqrt{y}} f(x,y)\,dxdy$

E şıkkı